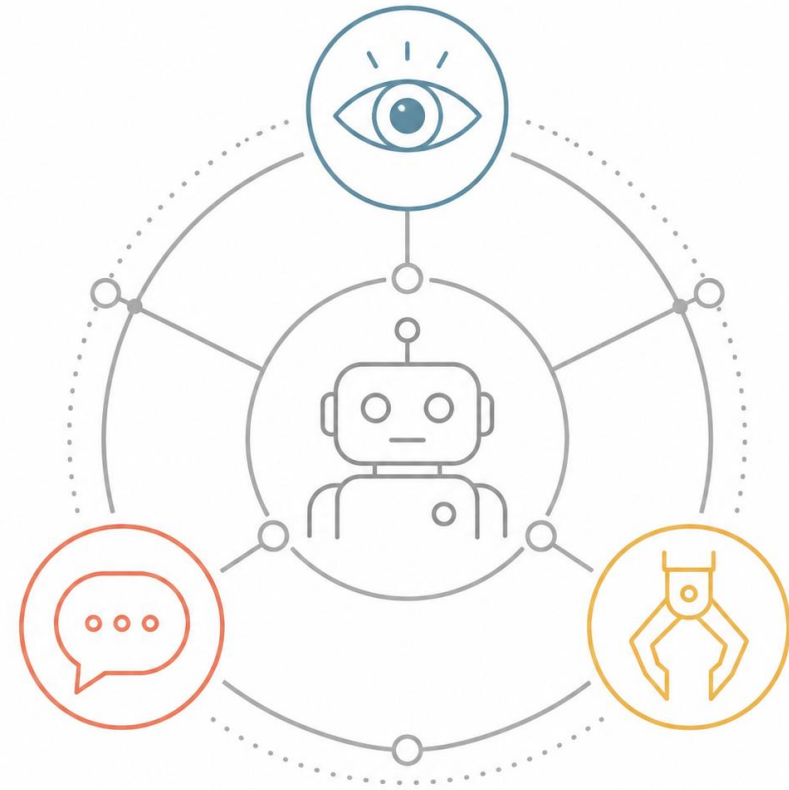


Vision-Language-Action (VLA) Models for Industrial Robotic Applications

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PhD Candidate, TU Berlin



Agenda

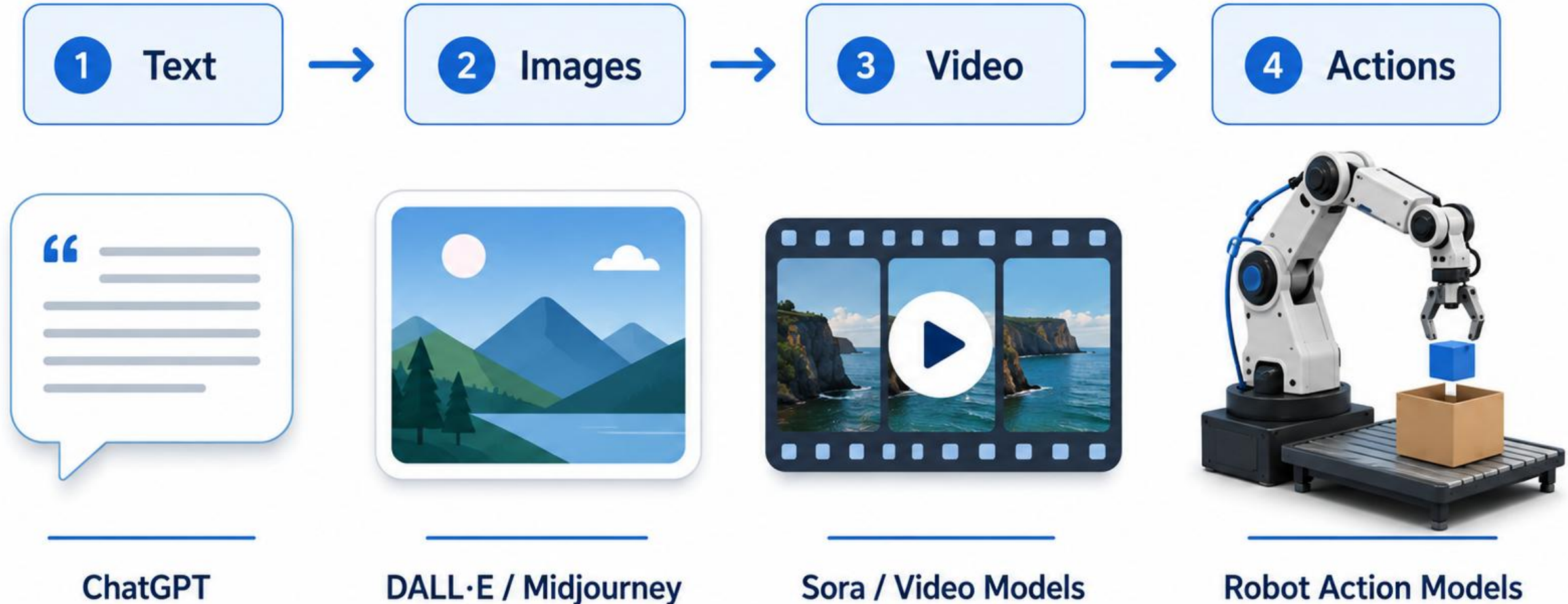
- **From Generative AI to Generative Action**
- **Vision-Language-Action Models**
- **Capabilities and Limitations**
- **Challenges in Industrial Scenarios**
- **Our Research**
- **Outlook & Takeaways**
- **Q&A + Discussion**



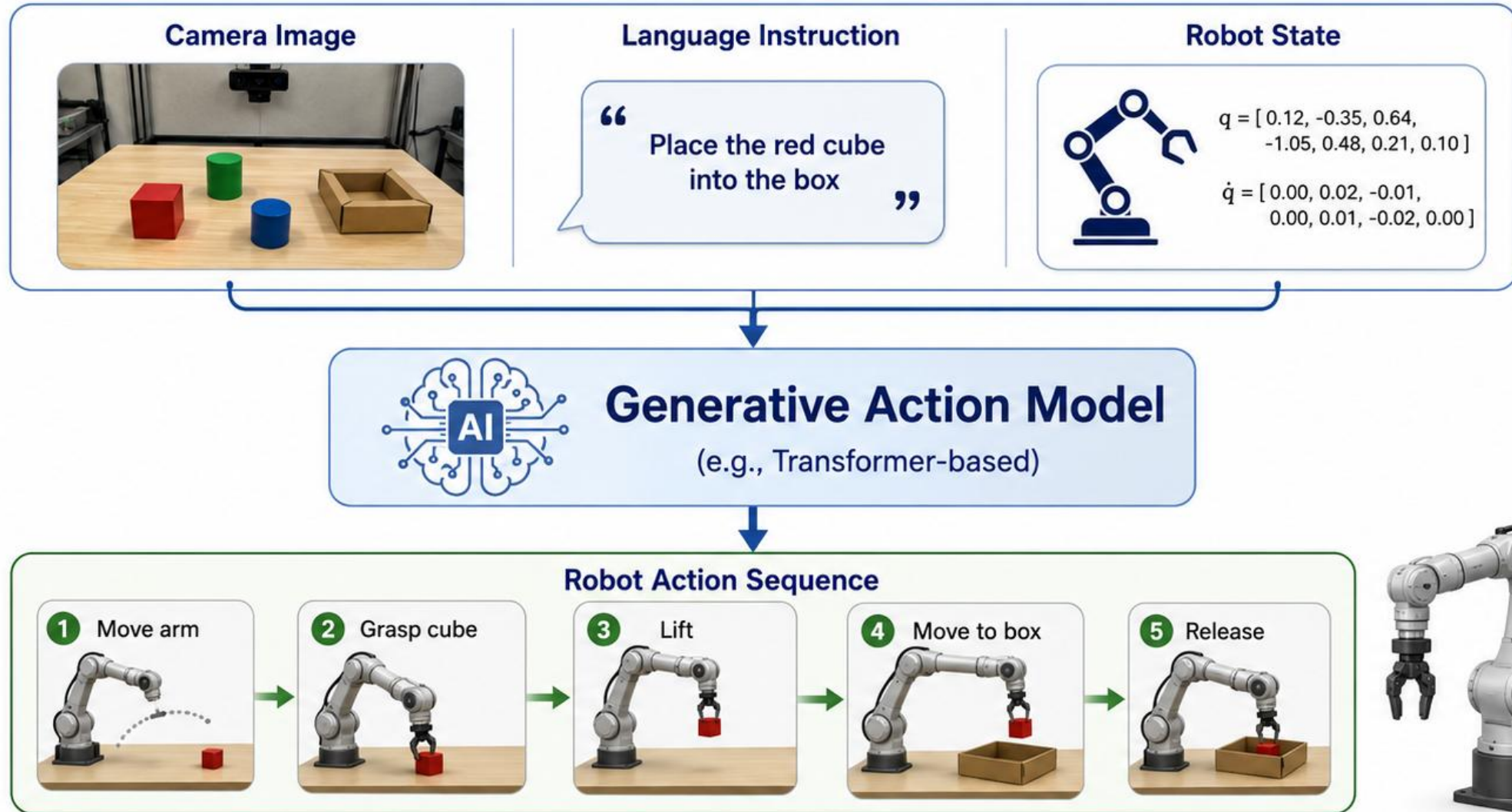
From Generative AI to Generative Action

A progression from understanding and creation to real-world execution

A horizontal progression:



Generative Action Models: The Basic Idea



The model generates an action trajectory conditioned on visual observations, language goals, and robot state.

Example: Task execution by a VLA

PROMPTS

Multi-stage Instructions

"Hi robot, can you make me a cheese, roast beef, and lettuce sandwich?"

Unseen Tasks

"Can you clean up only the trash, but not dishes?"

Situated Corrections

"That's not trash!"

User Constraints

"I'm allergic to pickles."

Open-ended Prompts

"It's movie night! Can you get me some chips, Oreos, and drinks?"

	Hi Robot	Hi Robot
	1. pick up bread 2. put bread on chopping board ...	
	1. pick up plastic container	
 "Oh sorry! I'll put it back"	1. open gripper 2. go higher	
 "Ok, I will avoid pickles"	1. pick up bread 2. put bread on chopping board	
	1. pick up orange chip bag 2. put chip bag in basket 3. pick up Oreo 4. put Oreo in basket	
		... 8. put lettuce on beef 9. pick up bread 10. put bread on lettuce
		2. put container in trash 3. pick up foil tray ...
		3. pick up bag of chips 4. throw away bag of chips ...
		3. pick up 1 slice of tomato 8. put bread on ham
		5. pick up water bottle ...

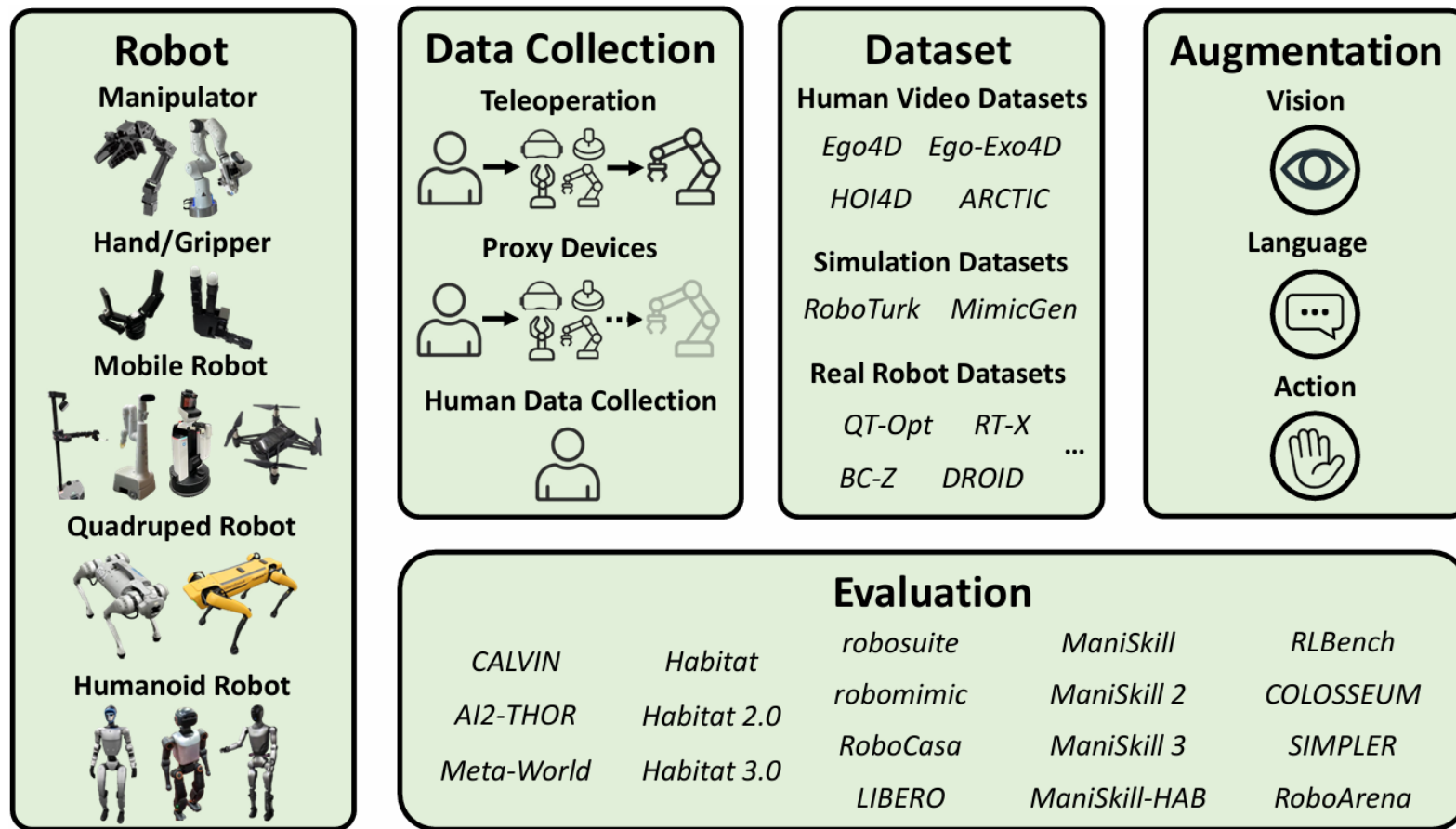
Video Examples:

<https://www.pi.website/research/hirobot>

Shi et al., Hi Robot: Open-Ended Instruction Following with Hierarchical Vision-Language-Action Models, Proceedings of ICML 2025, PMLR 267



What is a Vision-Language-Action Model?

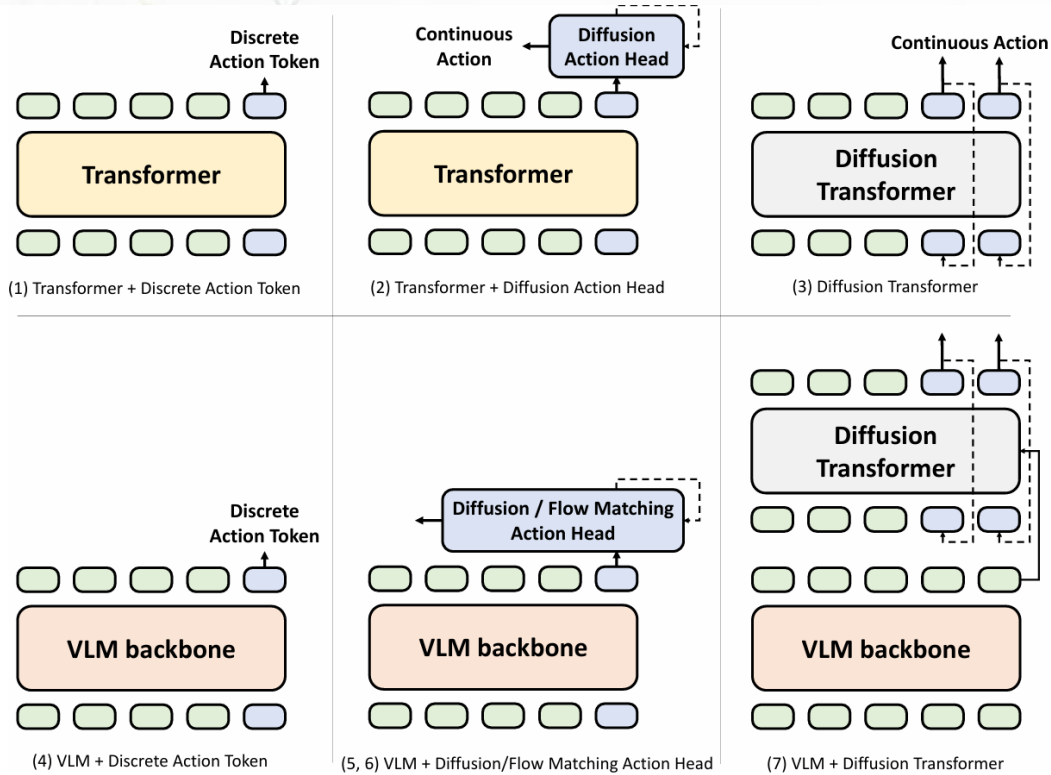


More data +
Larger models +
More diverse tasks
↓
Better representations
↓
Better generalization and
robustness
↓
Toward generalist robot
behavior

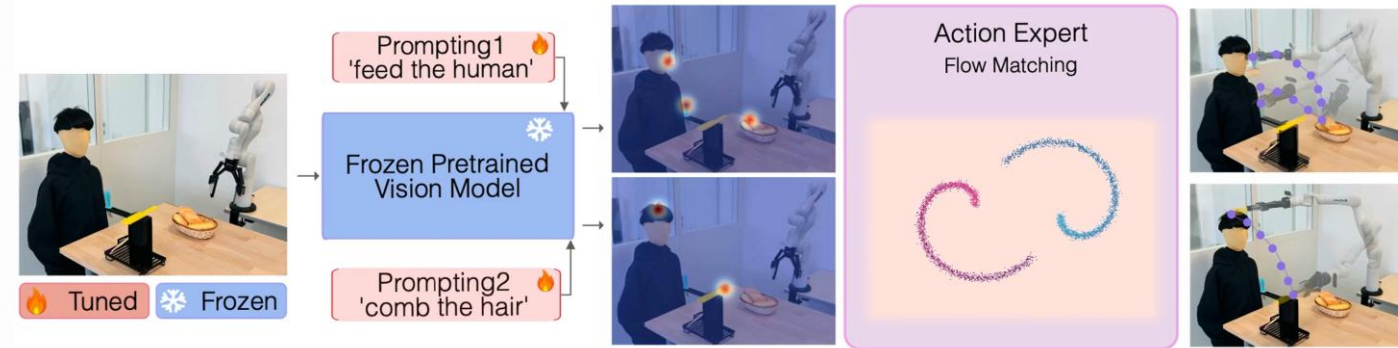
Kawaharazuka et. al, Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications, IEEE Access 2025



From Imitation Learning to Vision-Language-Action Models



VLA = imitation learning + vision-language understanding + generative action prediction



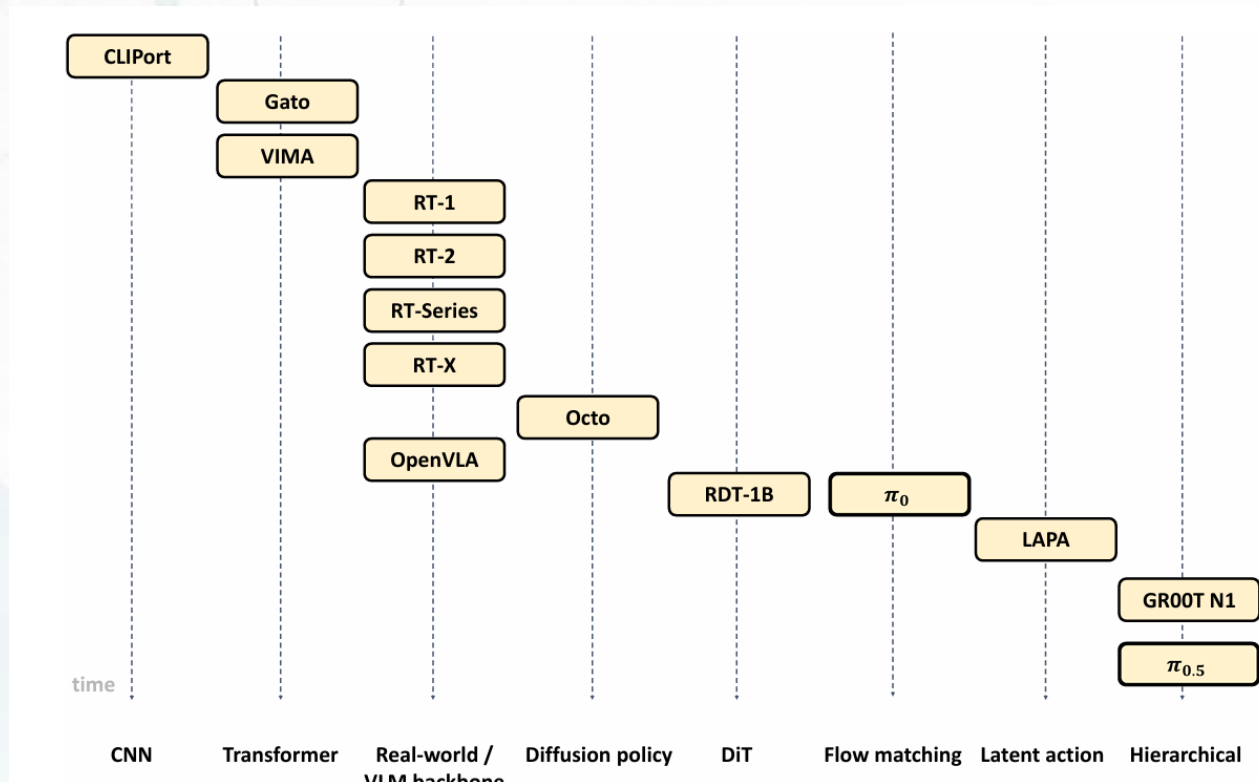
[Robot Manipulation with Flow Matching](#),

Zhang and Gienger, *Affordance-based Robot Manipulation with Flow Matching*, CoRR 2024

Kawaharazuka et al., *Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications*, IEEE Access 2025



Timeline of VLA Development

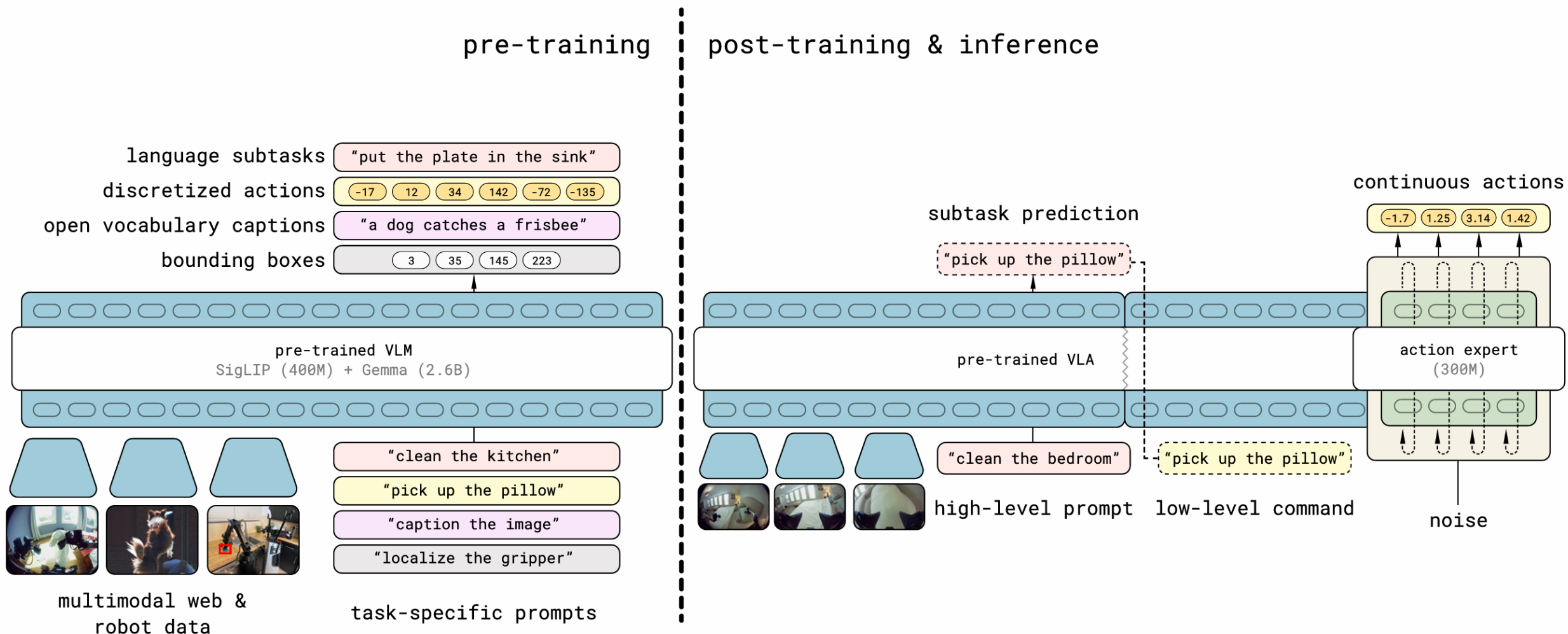


Kawaharazuka et al., *Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications*, IEEE Access 2025

Phase	Years	Main idea
Grounding and affordances	2021–2022	Connect language and vision to robot skills
Transformer robot policies	2022–2023	Scale imitation learning with large robot datasets
VLM-based VLAs	2023–2024	Use pretrained vision-language knowledge for robot action
Generative and deployable VLAs	2025–2026	Diffusion/flow actions, spatial grounding, real-time deployment, evaluation



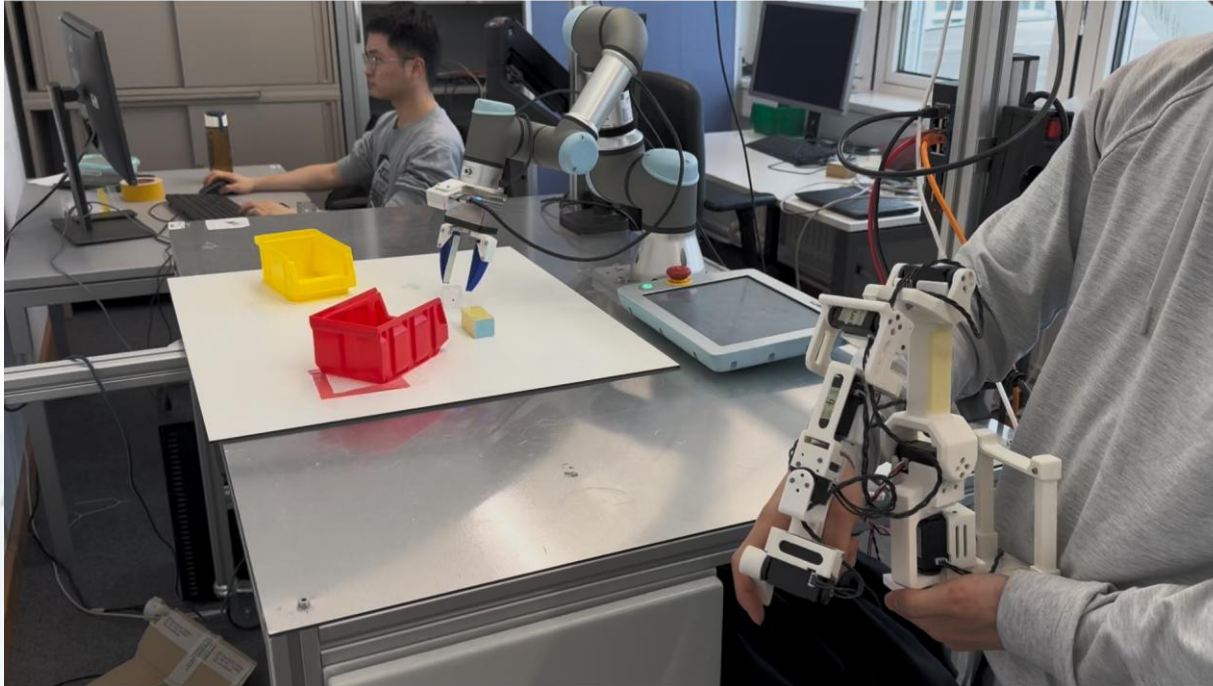
$\pi_{0.5}$: Model Architecture, Training and Inference



[Physical Intelligence, \$\pi_{0.5}\$: a Vision-Language-Action Model with Open-World Generalization](#)



Some Examples and Key Takeaways



Does the model work out of the box?

- No, finetuning is necessary

What data do they need?

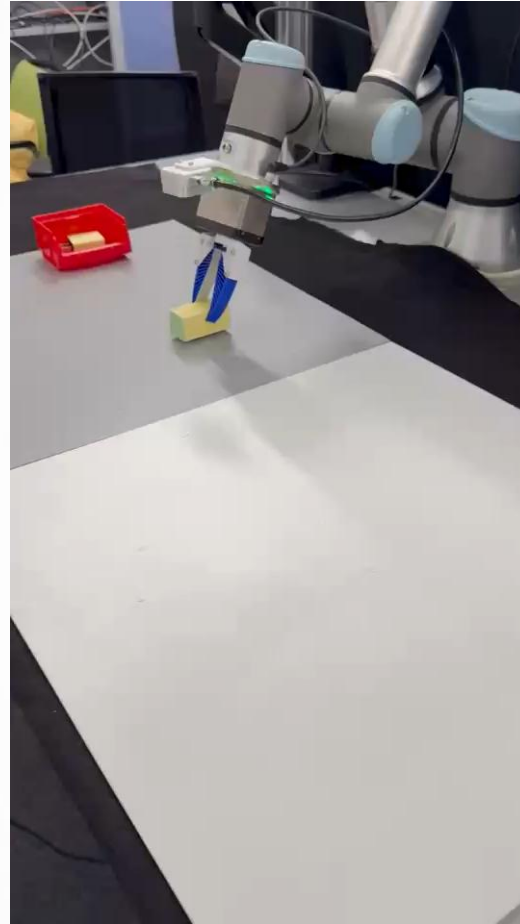
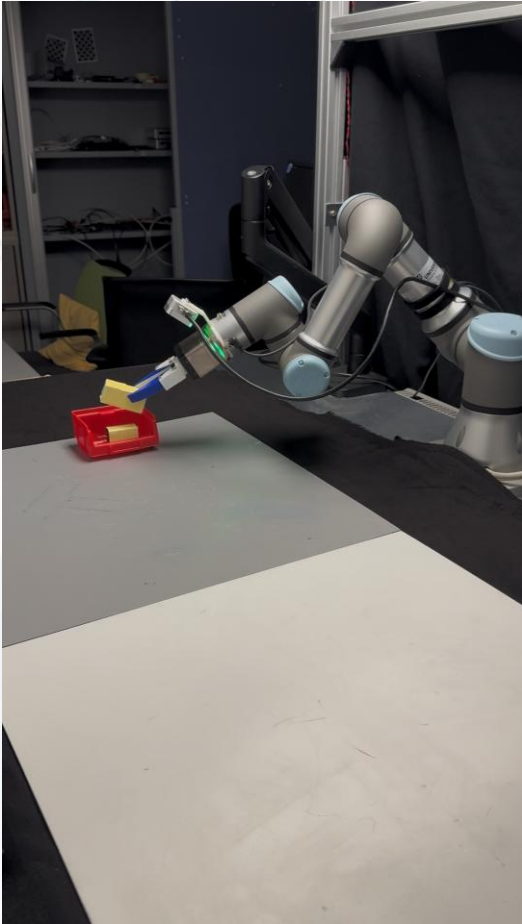
- Generally, teleoperation data to finetune the action head
- RGB data for the objects, visual surroundings etc.
- Language instruction connected with the actions

Can these be trained to perform multiple actions?

- Yes. More actions = More data = More compute



Some Examples and Key Takeaways



Does the Model perform well under changing conditions?

- Yes, generally, the model is robust to visual changes, object location changes and lighting conditions etc.

What are the 'emergent abilities' or advantages over standard end-to-end learning approaches?

- Self-corrective behaviour
- Can perform in novel settings
- Multi-step actions



Some Examples and Key Takeaways



What are the limitations or downsides?

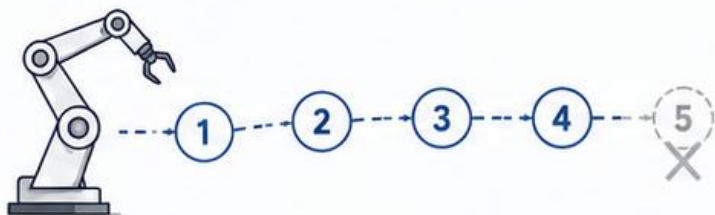
- End-to-end = Garbage in, Garbage out
- Autoregressive = Go from past to the future
- More challenging task = More data and compute needed
- Still poor at long-horizon tasks without guardrails
- Still poor at fine-tolerance tasks
- We are still restricted by the embodiment and the intelligence
- Not a panacea!



Current Limitations of Vision-Language-Action Models

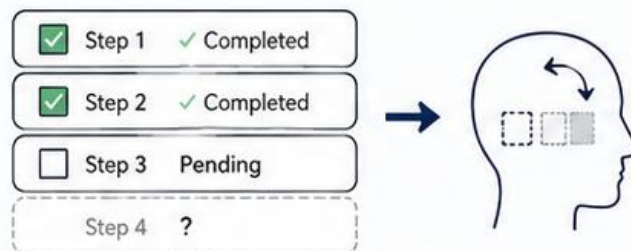
Why current VLA systems still struggle in real-world industrial workflows

1 Long-Horizon Execution



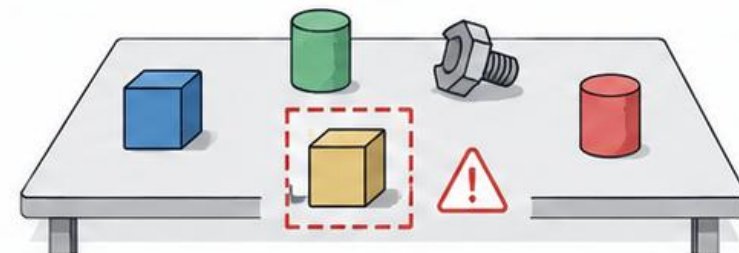
VLA systems can perform short skills, but often lose task progress over long multi-step procedures.

2 Weak Task Memory



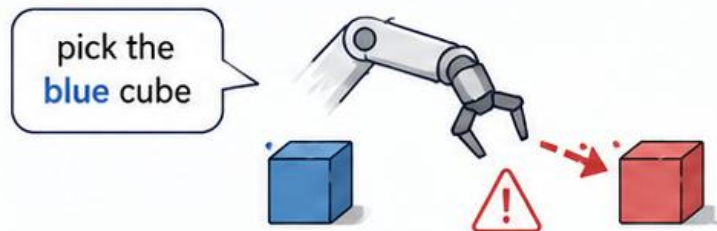
They may forget completed, pending, or failed steps without explicit task-state tracking.

3 Spatial Grounding Errors



Models may attend to the wrong object or region, especially in cluttered scenes.

4 Language Faithfulness



The robot does not always follow the instruction faithfully when multiple actions are possible.

5 Failure Detection & Recovery



Current VLA systems often lack reliable postcondition checks, retry logic, and recovery behavior.

6 Robustness & Evaluation

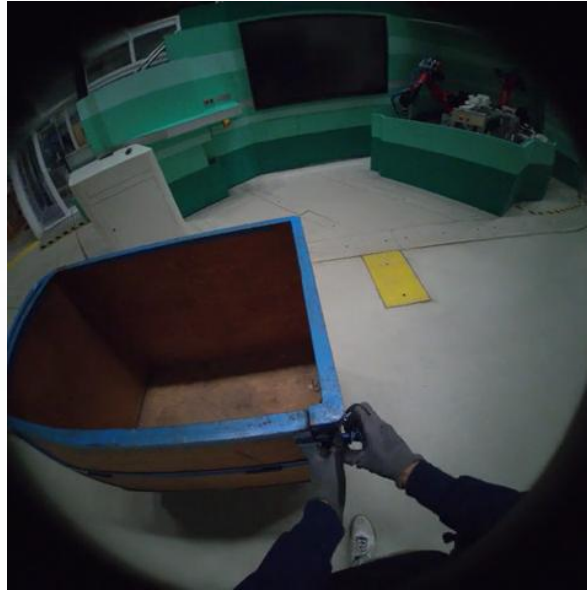


Success on benchmarks does not guarantee robustness, safety, or reliability under realistic disturbances.



The field is moving beyond bigger pretrained models toward **memory, grounding, controllability, robustness, and diagnostic evaluation.**

Egocentric-Exocentric Data & Benchmarks



- Egocentric data (w/ temporal annotations)
- Exocentric data
- Diverse industrial scenarios

IndEgo: A Dataset of Industrial Scenarios and Collaborative Work for Egocentric Assistants, NeurIPS 2025



Egocentric/Exocentric data, video/action understanding is highly relevant to VLAs and real-world robotics

(pretraining, **policy adaptation**, inference)

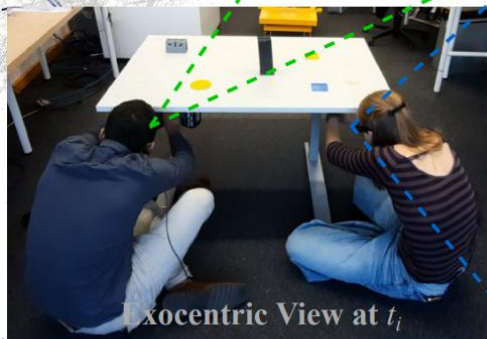
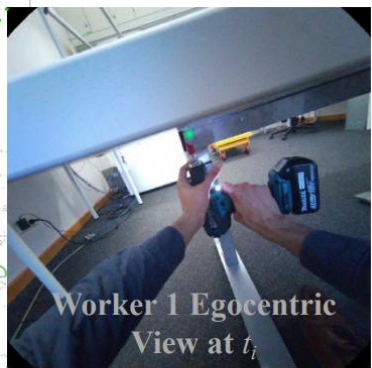
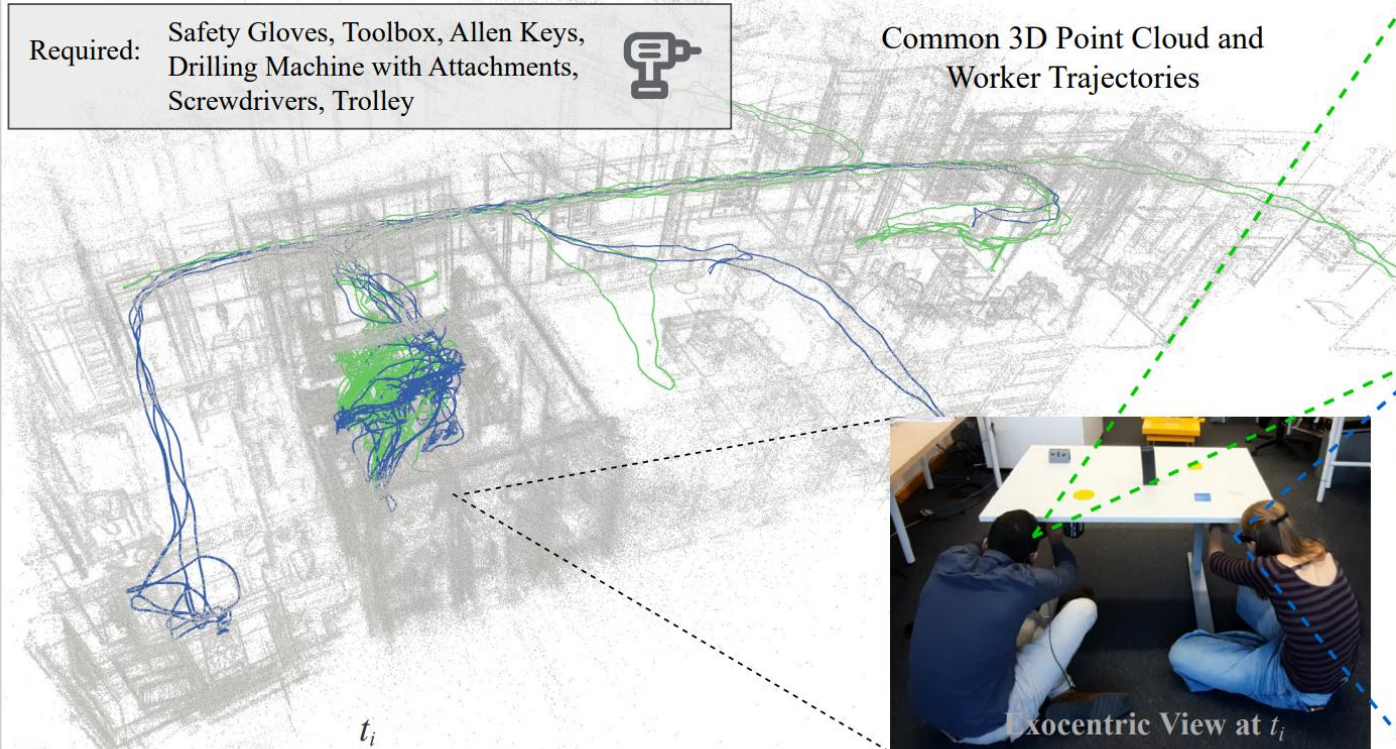


Structured Task Representations from Multimodal Data

Required: Safety Gloves, Toolbox, Allen Keys, Drilling Machine with Attachments, Screwdrivers, Trolley



Common 3D Point Cloud and Worker Trajectories




Worker 1

Open Toolbox

Attach Screw bit

Remove Bolt

Hand Tool to Coworker

Collect Bolts

Detach Tabletop

Place Top

Put on Gloves

Remove Crank

Sort Crank

Plan Action

Grab Tools

Remove Bolts

Grab Drill Machine

Attach new bit


Worker 2

Carry Tools

Grab Screwdriver

Remove Bolt

Take Drill Machine

Loosen bolts

Detach Tabletop

Sort Bolts

Put on Gloves

Read Instructions

Observe Coworker

Get Allen Keys

Remove Bolts

Detach Lower Plate

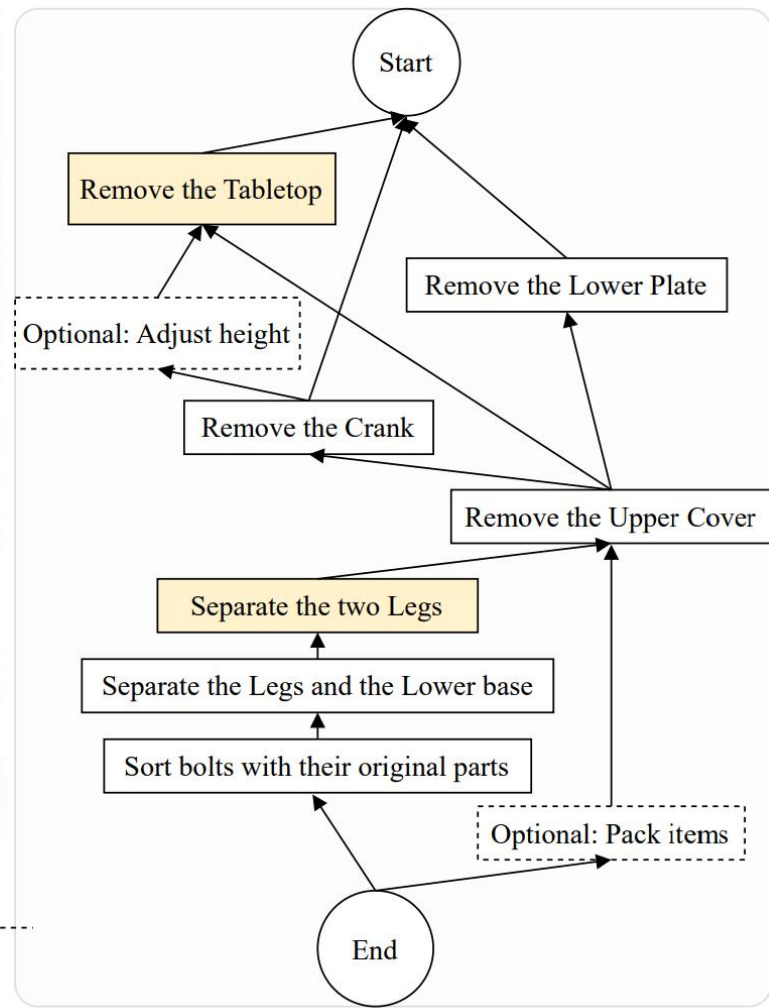
(Time >)

Remove the Tabletop

Remove Crank

Remove the Lower Plate

Task



IndEgo: A Dataset of Industrial Scenarios and Collaborative Work for Egocentric Assistants, NeurIPS 2025

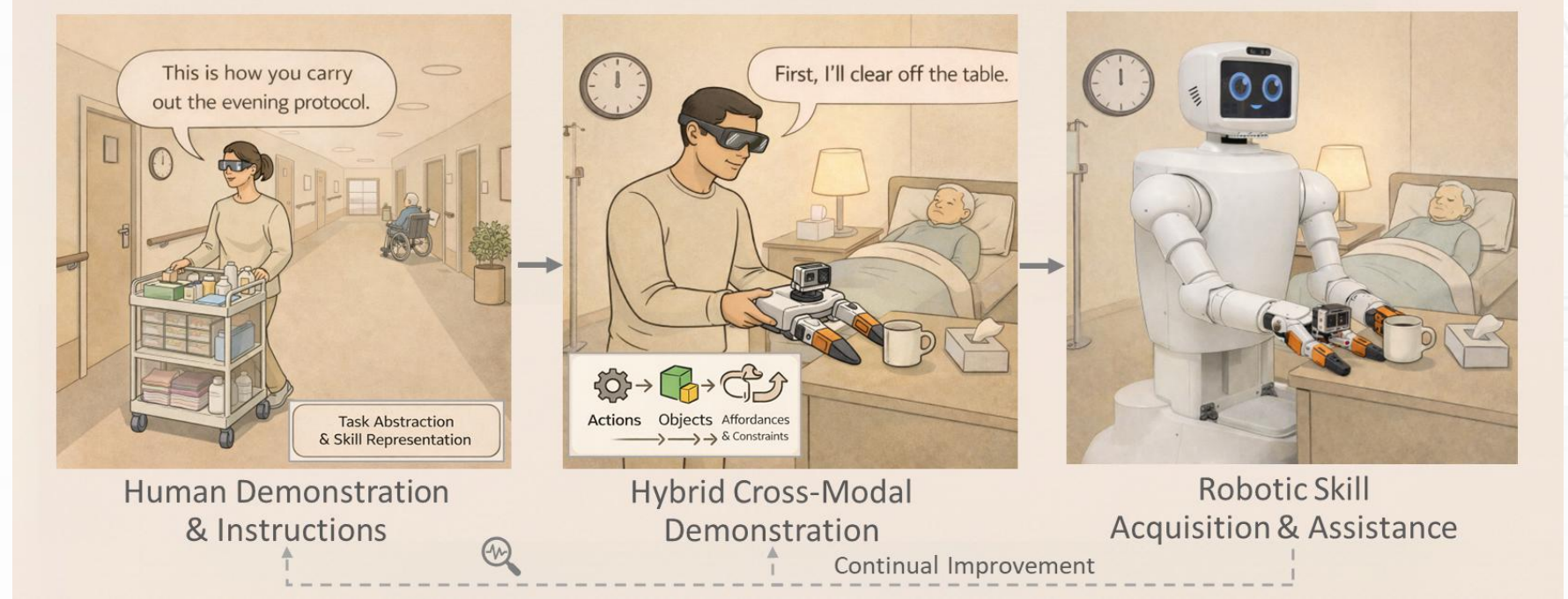


Extracting user intent and context from Ego/Exo Demonstration



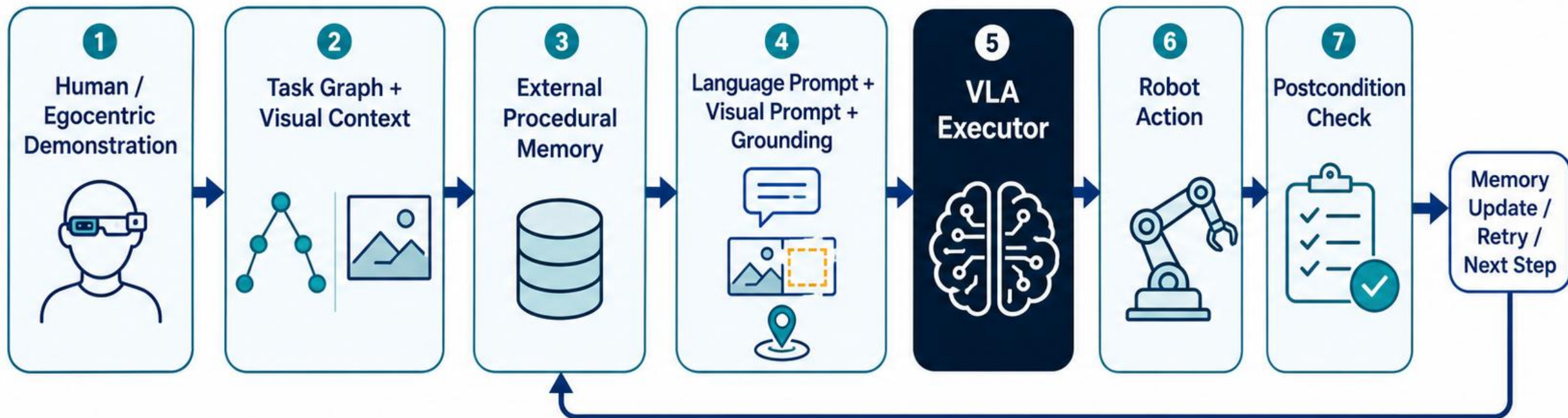
Hybrid Demonstration via a Universal Manipulation Interface

Chi et al., Universal Manipulation Interface: In-The-Wild Robot Teaching Without In-The-Wild Robots, RSS 2024



*The MIMIC Pipeline for Service Robotics.
(Left) Context-Aware Instruction.
(Centre) Hybrid Demonstration.
(Right) Embodied Execution*

MIMIC-VLA: Structured Context Around a VLA



- **Task graph:** defines valid steps and dependencies.



- **External memory:** tracks current, completed, pending, and failed steps.



- **Visual prompts:** highlight object, target region, or desired state.



- **Postcondition checks:** decide whether to continue, retry, or stop.

Why External Procedural Memory?

Internal memory helps the model remember. External memory helps the system control execution.

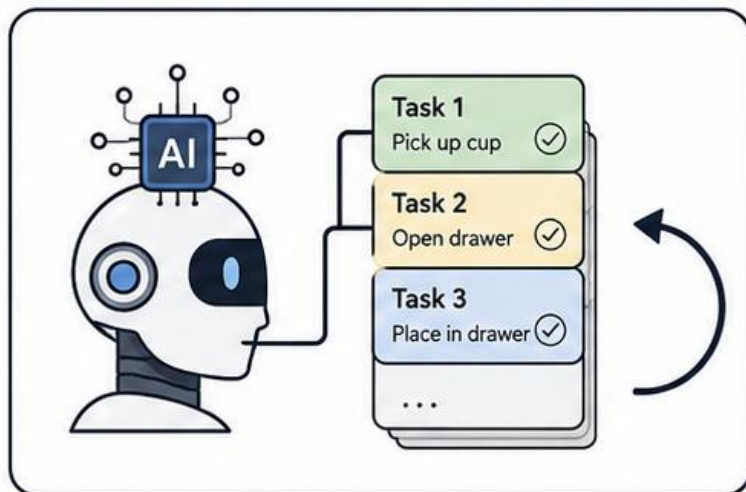
 Internal VLA Memory	 External Procedural Memory
Hidden inside the model	Explicit and inspectable
Learns temporal history	Tracks task state
Hard to edit	Human-editable
Helps perception/action continuity	Helps process control
Model decides what matters	Task graph defines what matters



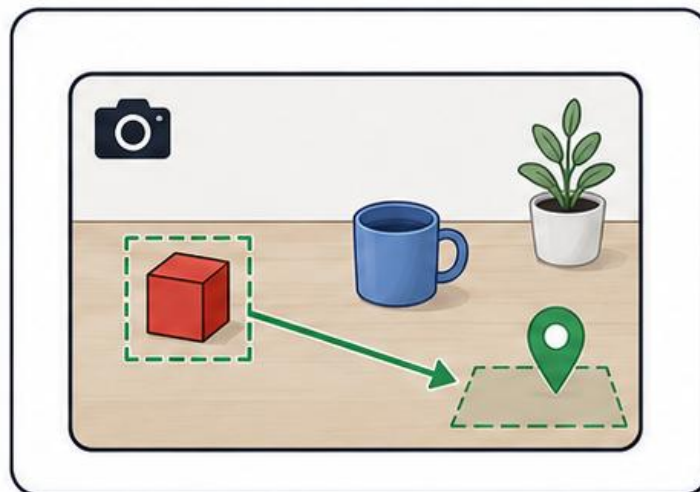
External memory is not just remembering; it is procedural control.



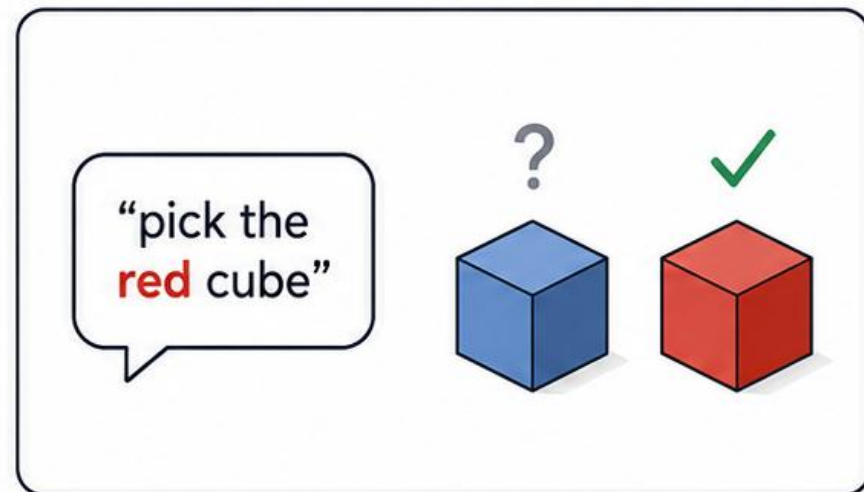
External memory acts as an execution-control layer: it selects the current step, chooses prompts, checks postconditions, and decides whether to continue, retry, or stop.



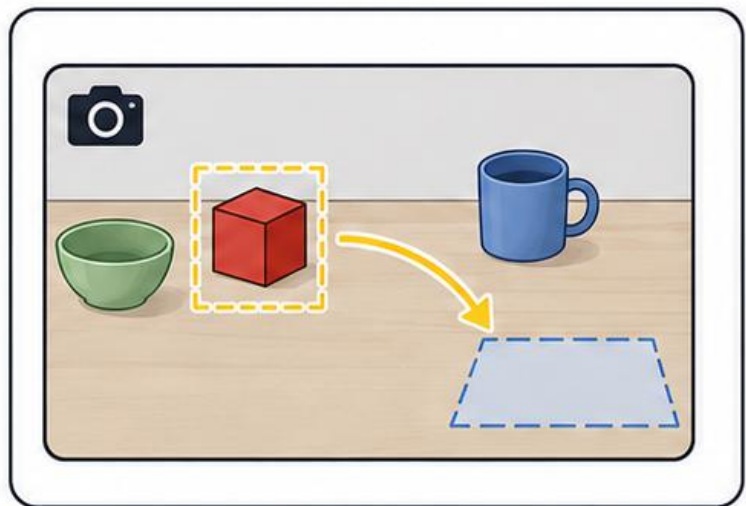
Memory



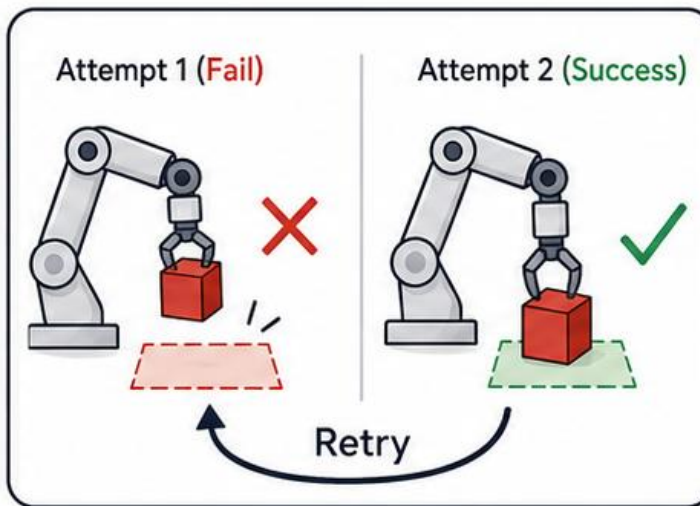
Spatial Grounding



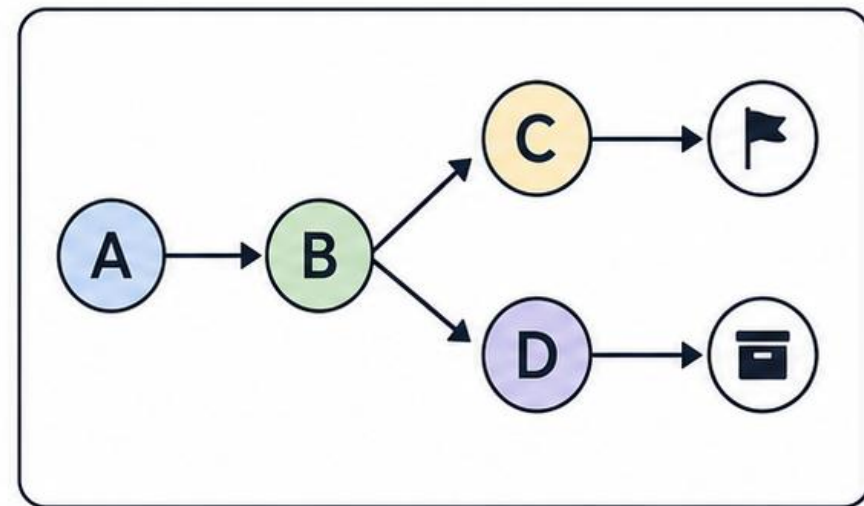
Language Faithfulness



Visual Prompting



Failure Detection



Task Graphs

Key Takeaways



- 1** VLAs connect perception, language, and action



Generate robot actions from images, instructions, and robot state.

- 2** Robot foundation models enable broader generalization



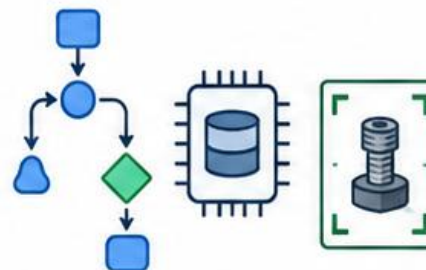
Larger models and broader data support more general robot behavior.

- 3** Manufacturing needs more than isolated skills



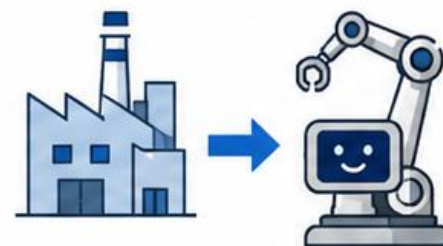
Industrial tasks require memory, task structure, grounding, and recovery.

- 4** Our research adds structure around the VLA



External procedural memory and visual prompts guide what the robot should do next.

- 5** Long-term vision: reliable industrial robot workflows



From isolated robot skills to reliable, structured industrial workflows.



The future is not only robots that can act, but robots that can follow structured procedures reliably.